

ELECTRONIC METRONOME

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A metronome is used by musicians for practice in maintaining a consistent tempo, or rubato, around a fixed beat. This circuit produces a regular beat at the rate of 40 to 200 beats per minute. It accentuates every second, third, fourth, fifth, sixth or eighth beat, which is adjustable as per your liking and requirement. Every beat is indicated by the glowing of an LED. The accented beat is indicated by another LED.

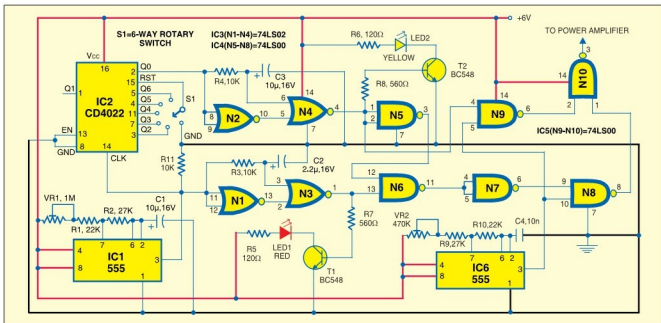
IC1 drives the pulse generator. The length of the pulse is about 10 ms, and it appears at pin 1 of IC3 (NOR gate N3). At each pulse, the red LED (LED1) flashes to indicate occurrence of the beat. The pulse passes through NAND gates N6 and N7 of IC4. The pulse output from pin 6 of N7 is fed to NAND gate N8. The audio signal output generated by another multivibrator (IC6) is also fed to gate N8. The audio signal can be adjusted to obtain a note of suitable pitch.

The output from IC1 also goes

is longer (about 40ms) and is used to mark the accented beat. The result is a 'tick' sound lasting about 40 ms, which sounds every second, third, fourth, fifth, sixth or eighth beat, depending on the setting of S1. The accent pulse makes the yellow LED (LED2) flash.

It is important that the base 'tick' note or beat is not heard on the accented beat. This is achieved by gates N5 through N7 of IC4.

The final audio signal appears at pin 3 of IC5 (NAND gate N10). This



The beat is derived from an astable multivibrator (IC1) running between 0.67 Hz (40 beats per minute) and 3.47 Hz (208 beats per minute), and a pulse generator built around NOR gates N1 and N3, resistor R3 and capacitor C2. The beat covers all the musical *tempi* from *adagio* to *presto*. The results are a very short burst of sound, reminiscent of the 'tick' of a mechanical metronome. If you prefer a beep rather than a tick sound, the pulses should be lengthened by reducing the value of R3 to, say, 5.6 or 6.8 kilo-ohms.

to IC2 (CD4022), which is a divide-by-eight counter/divider with eight decoded outputs. Rotary switch S1 allows the counter to be reset every two, three, four, five or six counts, or cycle through eight counts without resetting.

Output Q0 of IC2 drives the second pulse generator built around NOR gates N2 and N4, resistor R4 and capacitor C3. The output is an accented beat pulse, which is fed to NAND gates N5 and N9 and the base of transistor T2. Since C3 has a higher capacitance than C2, this pulse

signal can be fed to the audio power amplifier stage. When you supply 6V DC to the circuit, you can hear the base or tempo beats and accented beats from the speaker of your power amplifier. The red LED (LED1) flashes with the beat and the yellow LED (LED2) flashes on the accented beat.

Construction and testing is simple. Assemble the circuit on a breadboard or general-purpose PCB. Mount all the components, except S1, and temporarily connect pin 15 of IC2 to ground rail. IC1 produces an audible tick sound